

TAO SHEN

Ph.D. Candidate, National University of Singapore

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EDUCATION

National University of Singapore (NUS)

Jan 2023 - Present

Ph.D. Candidate in Statistics

- Department: Department of Statistics and Data Science
- Advisors: Dr. Yifan Cui, Dr. Wanjie Wang

National University of Singapore (NUS)

Aug 2021 - Dec 2022

Master of Science in Statistics

- Department: Department of Statistics and Data Science

The Chinese University of Hong Kong, Shenzhen (CUHK-SZ)

Sept 2017 - Jun 2021

Bachelor of Science in Statistical Science

- Department: School of Data Science (SDS)
- With Honours, First Class

PUBLICATIONS

Articles Published

Shen, T. and Wang, W. (2026). Optimal Network-Guided Covariate Selection for High-Dimensional Data Integration. Submitted to *Journal of the American Statistical Association*.

Shen, T. and Cui, Y. (2025). Proxy-Aided Demand Learning with an Application to Various Pricing Problems. *Operations Research*.

Zhang, B., Shen, T., and Cui, Y. (2025). Neural Estimation of Treatment Bridge Functions for Proximal Causal Inference. *Statistical Analysis and Data Mining: An ASA Data Science Journal*, 18(5), e70045.

Shen, T. and Cui, Y. (2024). Optimal Treatment Regimes for Proximal Causal Learning. *Advances in Neural Information Processing Systems*.

Shen, T. and Cui, Y. (2024). Statistical Reinforcement Learning and Dynamic Treatment Regimes. In *Statistics in Precision Health: Theory, Methods and Applications*, pp. 163–200.

Articles Submitted

Cui, Y., Liu, J., Shen, T., Qi, Z., and Chen, X. (2026). Learning Robust Treatment Rules for Censored Data. Minor revision, *Journal of the American Statistical Association*.

Preprints

Wang, W. and Shen, T. (2026). NetPTR: Optimal Differentially Private Spectral Community Detection on Sparse Networks. arXiv:2606.26145.

Shen, T., Tong, X. T., and Wang, W. (2026). Efficient Propose-Test-Release for Optimal Differentially Private Estimation. arXiv:2605.03264.

Shen, T. and Wang, W. (2026). Community Detection on Inhomogeneous Multilayer Networks with Extreme Sparsity. arXiv:2603.22838.

TEACHING

Tutor for DSA3362: Predictive Data Analytics	<i>Spring 2026</i>
Student Helper for ST5188: Advanced Data Science Project	<i>Fall 2025</i>
Tutor for ST3131: Applied Regression Analysis	<i>Spring 2025</i>
Student Helper for ST5188: Advanced Data Science Project	<i>Spring 2025</i>
Student Helper for ST5188: Advanced Data Science Project	<i>Fall 2024</i>
Tutor for ST2334: Probability and Statistics	<i>Spring 2024</i>
Student Helper for ST5188: Statistical Research Project	<i>Spring 2024</i>
Student Helper for ST5188: Statistical Research Project	<i>Fall 2023</i>

INVITED TALKS

The ePTR Framework: Differential Privacy through Stability Certification	<i>Jul 2026, Singapore</i>
• Workshop on Modern Challenges in Data Decentralization	
Learning Robust Treatment Rules for Censored Data	<i>Aug 2025, Online</i>
• 8th International Conference on Econometrics and Statistics (EcoSta 2025)	
Community Detection in Multilayer Networks with Complex Inhomogeneity	<i>Jun 2025, Tokyo</i>
• The 2025 Workshop on Statistical Network Analysis and Beyond (SNAB 2025)	
Proxy-aided Demand Learning with An Application on Various Pricing Problems	<i>Jul 2024, Beijing</i>
• 7th International Conference on Econometrics and Statistics (EcoSta 2024)	
Proxy-aided Demand Learning with An Application on Various Pricing Problems	<i>Jun 2024, Wuhan</i>
• International Chinese Statistical Association China Conference 2024 (ICSA 2024)	
Optimal Individualized Decision-Making with Proxies	<i>Jun 2023, Chengdu</i>
• International Chinese Statistical Association China Conference 2023 (ICSA 2023)	
Learning Optimal Treatment Regime with Proxies	<i>Dec 2022, London</i>
• 15th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2022)	

ACHIEVEMENTS

NUS Faculty of Science Best Graduate Researcher Award	<i>Year 2026</i>
Faculty of Science PhD Conference Award (FPCA)	<i>Year 2025</i>
NUS Research Scholarship	<i>Year 2023-2026</i>
Dean's List Award of School of Data Science	<i>Year 2019-2021</i>
Academic Performance Scholarship of School of Data Science	<i>Year 2019-2020</i>
Master's List Award of Shaw College	<i>Year 2019-2020</i>

PROFESSIONAL SERVICE

Journal Referee for:
• Journal of Business & Economic Statistics
• Journal of Computational and Graphical Statistics
• Biostatistics & Epidemiology
• Biometrical Journal
• American Journal of Epidemiology
Conference Reviewer for:
• NeurIPS